

Off-Grid Carbon-Neutral INTEGRATED EV CHARGING COMPLEX

The proposed off-grid integrated system will be very useful when there is a sudden grid failure. It will continue to provide charging for two- and four-wheelers. Besides it will enable bio-CNG dispensing, sale of milk, sale of organic manure, sale of cow urine for medical usage that can even be exported, and let you claim carbon credits as well



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Many new technologies are getting introduced across the length and breadth of our country, such as:

- EV charging stations
- Compressed biogas plants
- Agri-voltaic farms (coming soon)
- Battery based energy storage systems (coming soon)
- Hydrogen generation systems (coming soon)

As of today, all these systems are being planned in silos and are being built as independent systems. But this is the right time to plan integrated systems, which can offer additional benefits, apart from their intended use. It will allow us to derive multiple benefits. A brief introduction to each of the above listed technologies is given below.

shall focus on the charging stations being installed along the highways and major roads between cities.

Almost all of these charging stations are grid based. Grid connected systems are easy to install, need less investment, and take up less space. However, they are going to put huge load on the already overloaded grid. Also, imagine what will happen if there is a grid failure?

Therefore, to tackle such a situation, we should have some off-grid charging stations as well. Such charging stations will reduce load on the grid and will be very beneficial in case of a prolonged grid failure.

There are two- and three-wheeler chargers as well as four-wheeler chargers. The four-wheeler chargers themselves are of three types.

Level 1 (L1) chargers. These are slow chargers that take eight to ten hours to charge a depleted battery. Hence these are used in residences for overnight charging.

Level 2 (L2) chargers. These take about four hours to charge. Such chargers are installed in housing complexes, workplaces, restaurants, malls, etc.

Level 3 (L3) chargers. These are DC superfast chargers. An L3 charger can charge a depleted battery to 80% within an hour or two. These are special chargers that are installed by the automotive manufacturers or government agencies along the highway.

Compressed biogas plants

Biogas plants are quite common. These plants use bio waste material, cow dung,

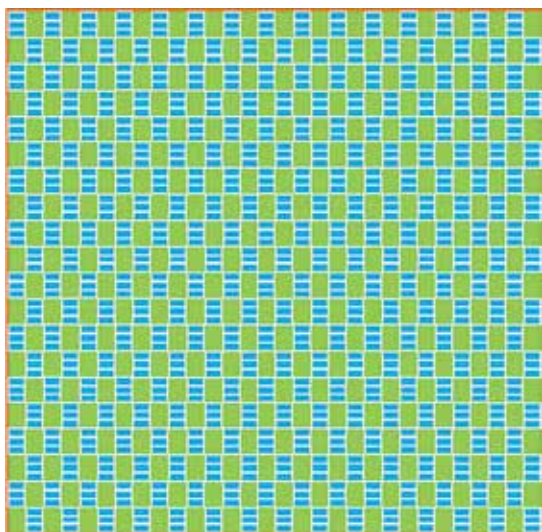


Fig. 1: Top view of Agri-voltaics farm spread over one acre land with 900 panels mounted in alternate slots

EV charging stations

The number of electric vehicles is increasing day by day, which is also increasing the need for EV charging stations. Both government and private companies have started building charging stations. These stations are of two main categories: within the city and along the highways. Here we